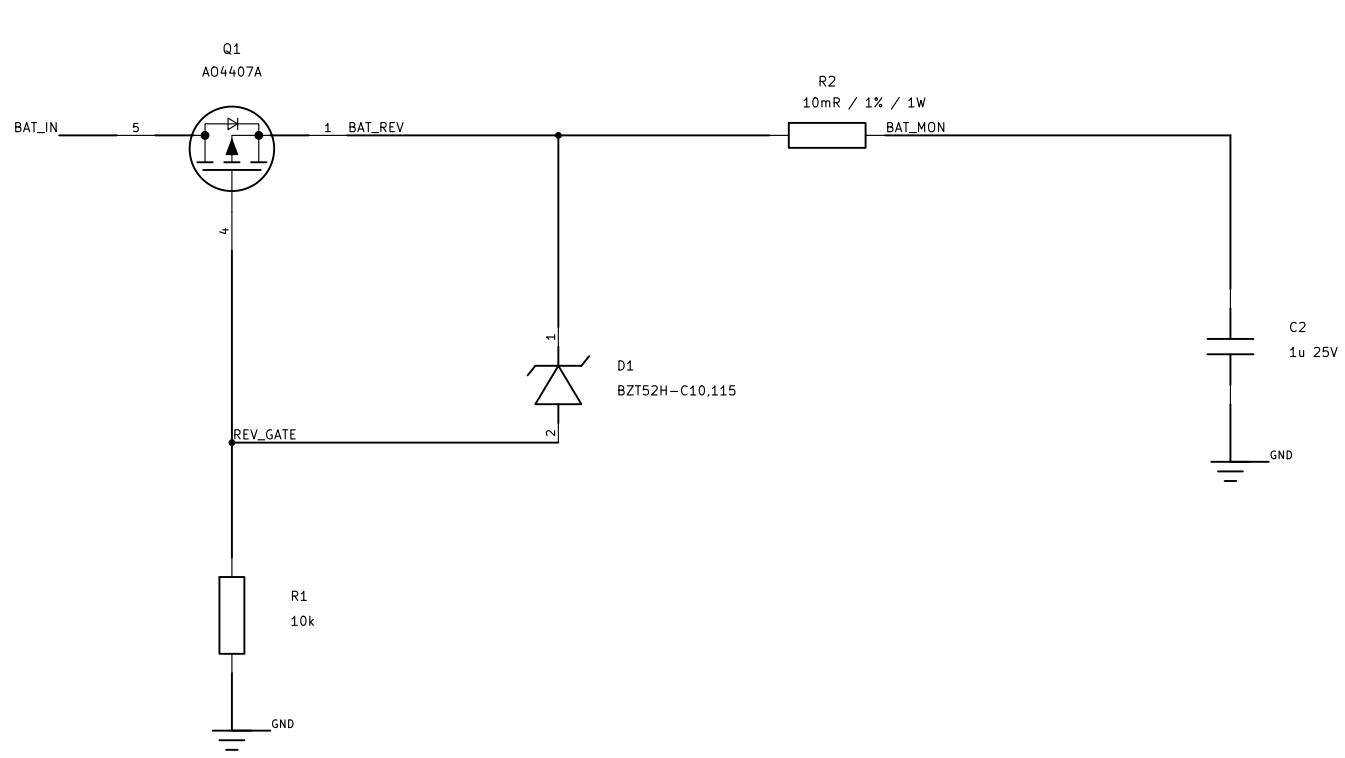


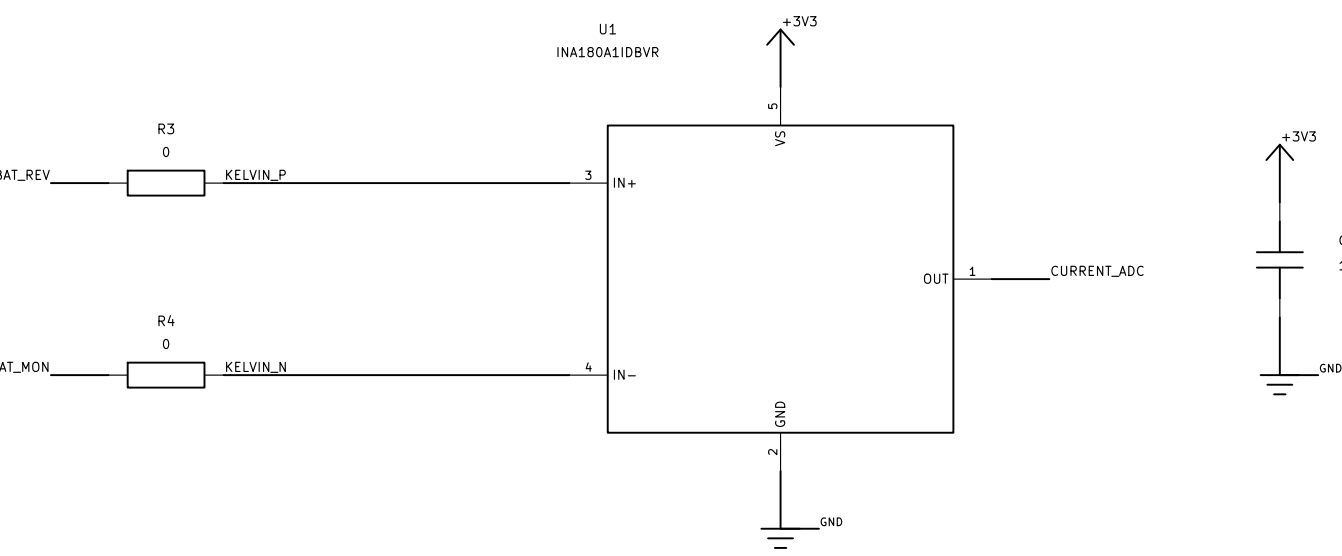
Power flows left to right. Supplies above, returns below. PROTOTYPE / UNVALIDATED.

01 Pack input / reverse protection / shunt

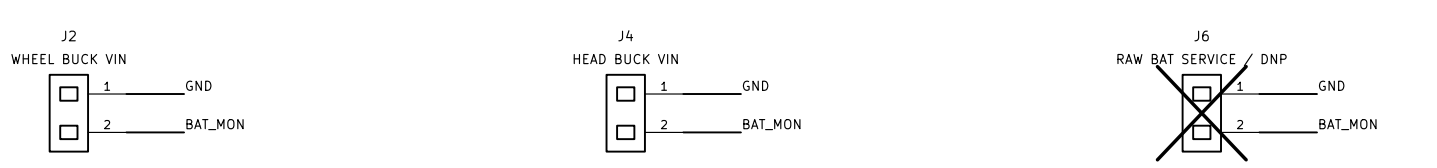


Q1's master stage provides BAT_IN, no charger on this PCB.

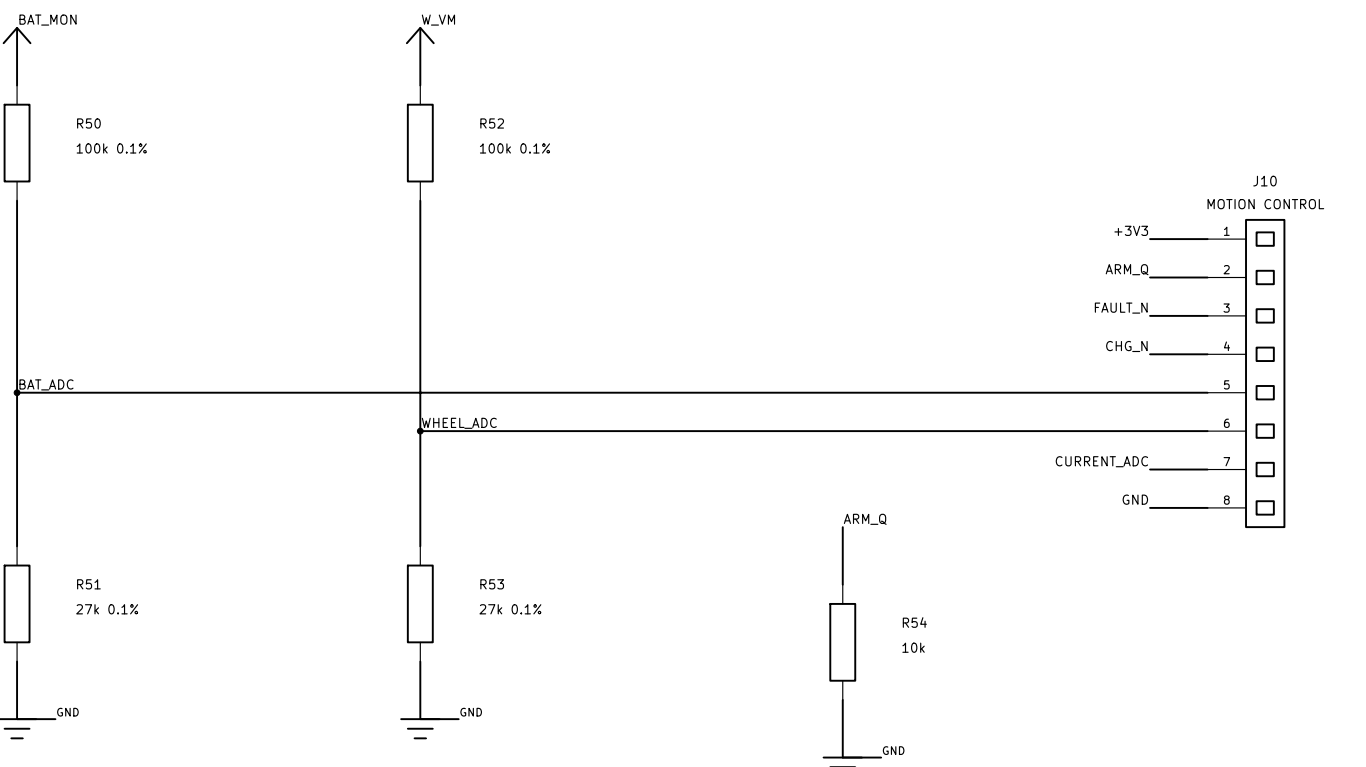
02 Kelvin current sense



03 External converter input distribution

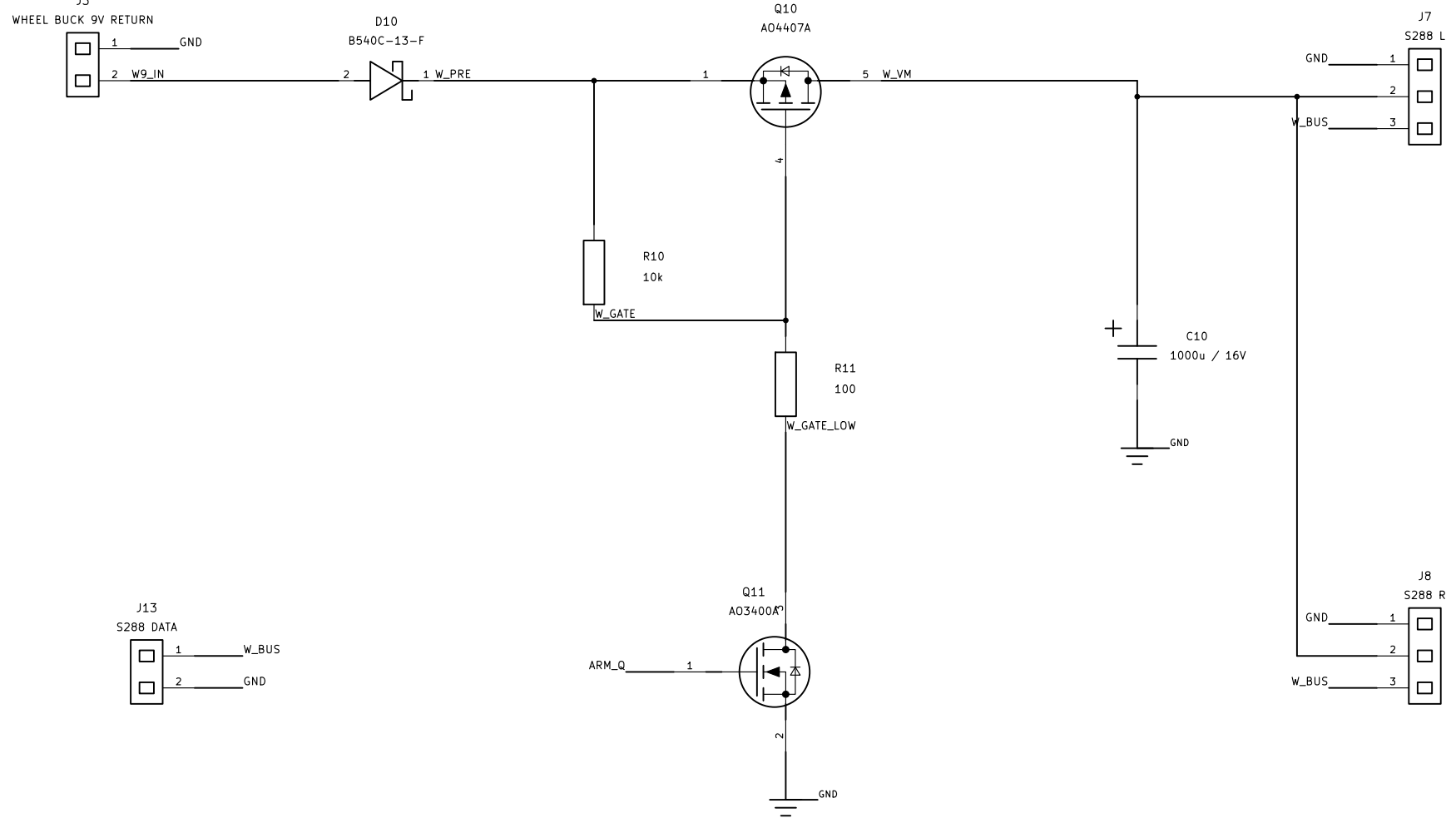


04 Voltage monitor / motion connector



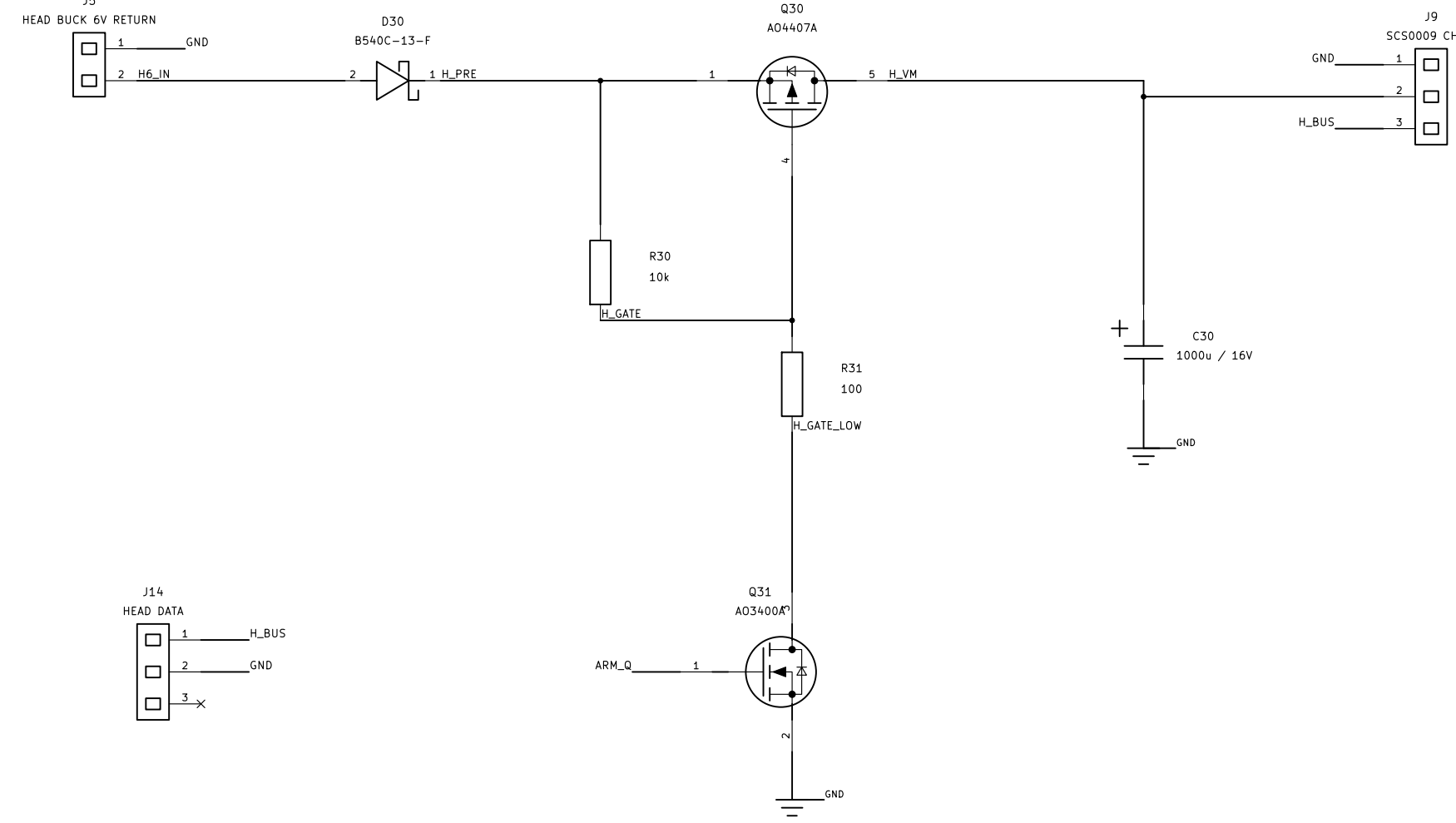
Divides ratio 27 / 127. CurrentSense comes from block 02.

05 Wheel 9 V conditioning



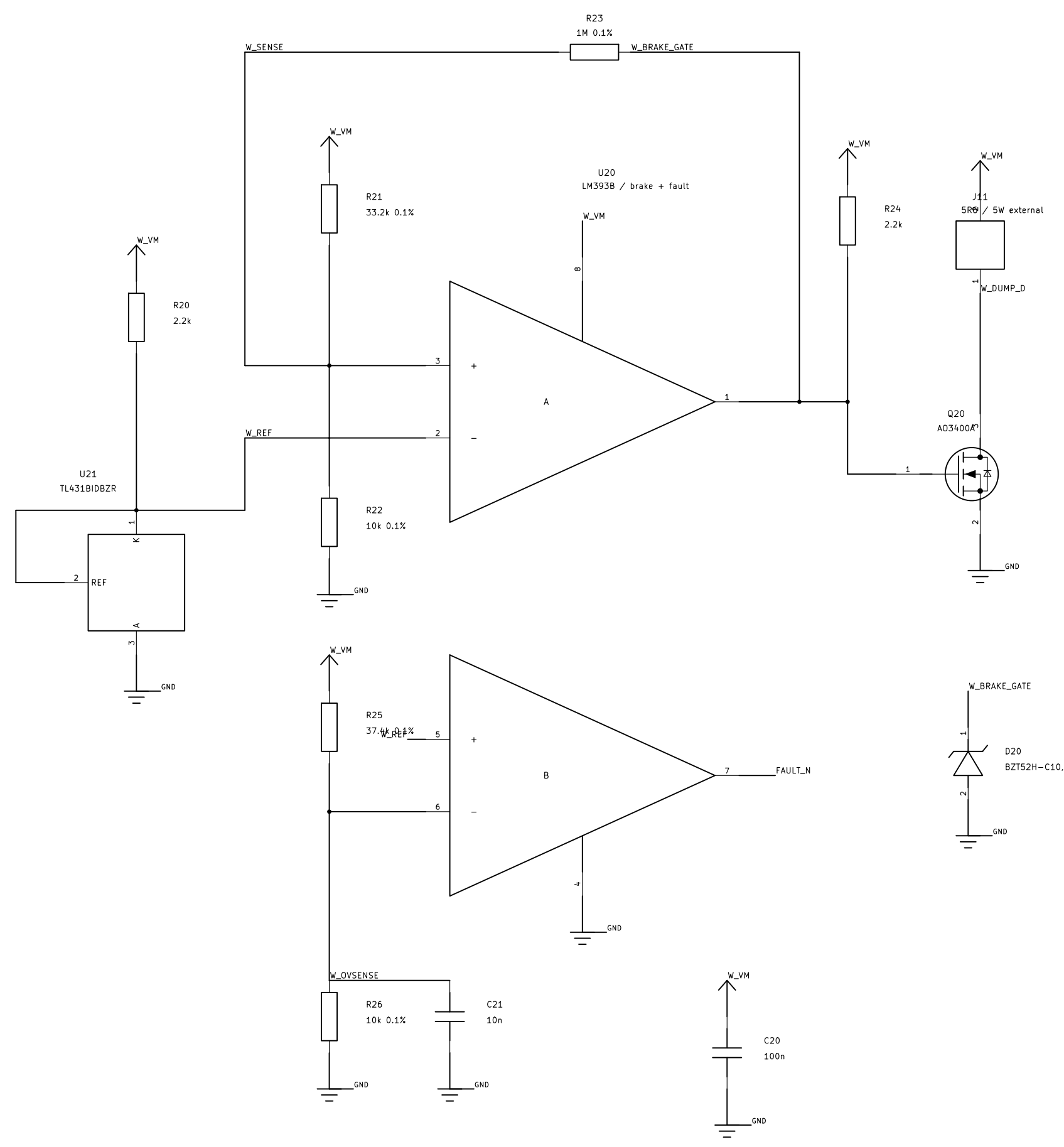
External back return, touch, brake energy and thermal performance. NOT TESTED.

06 Head 6 V conditioning



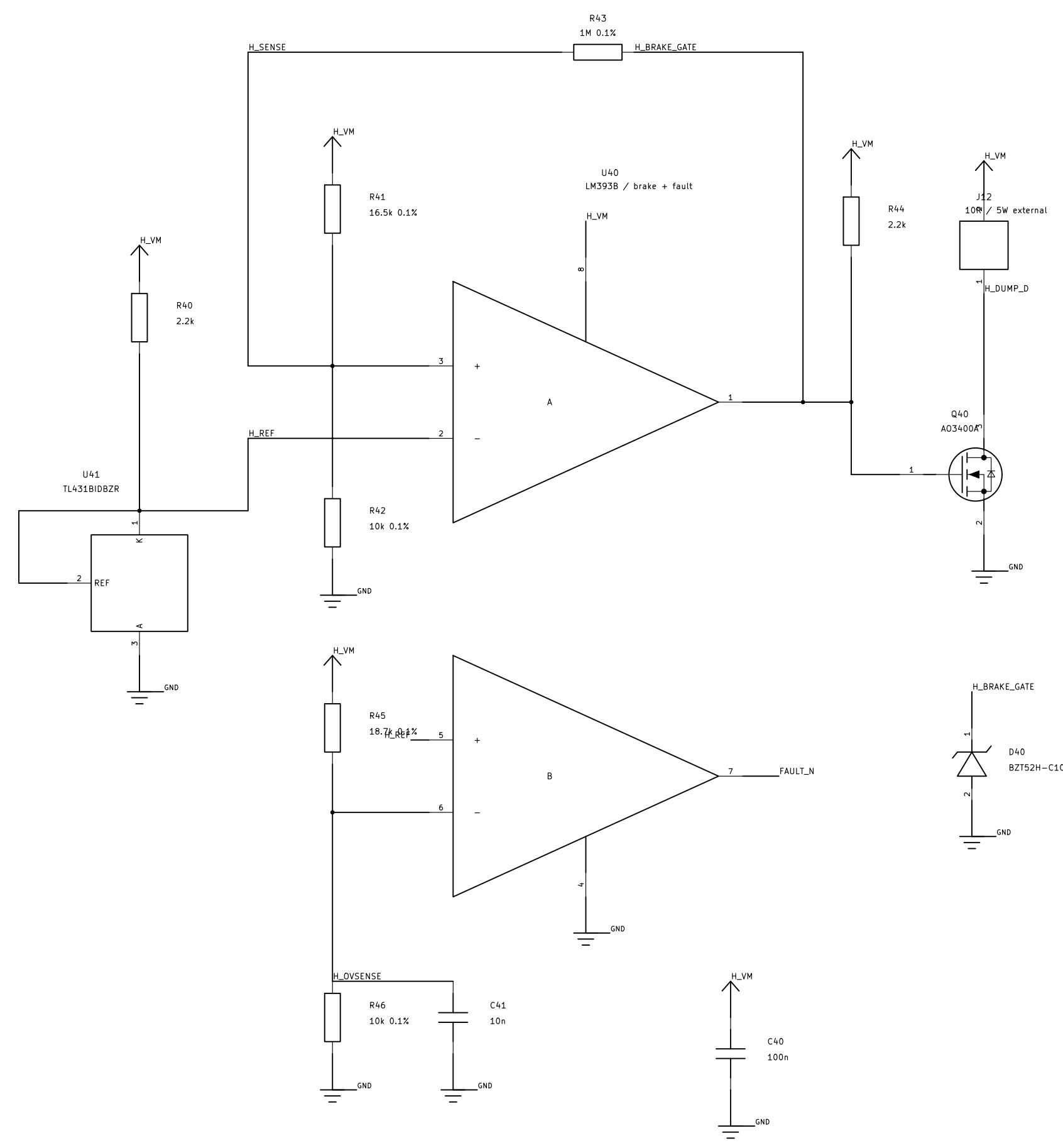
External back return, touch, brake energy and thermal performance. NOT TESTED.

07 Wheel rail / brake and overvoltage



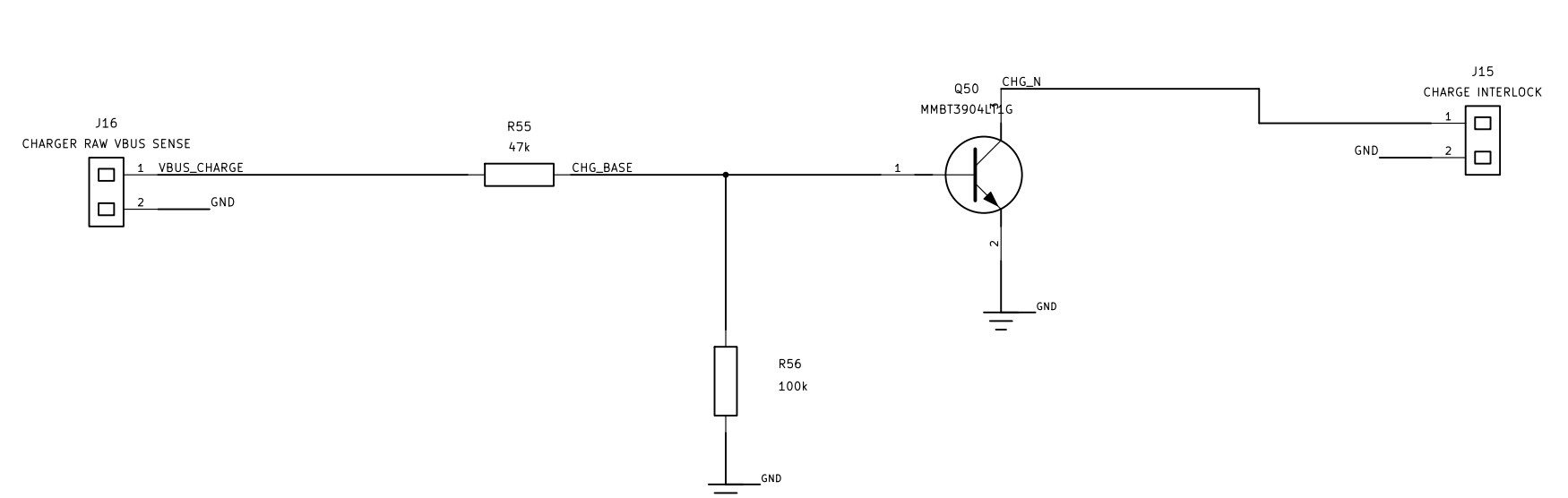
Two comparators share one J203B. External dump resistor needs prior qualification.

08 Head rail / brake and overvoltage



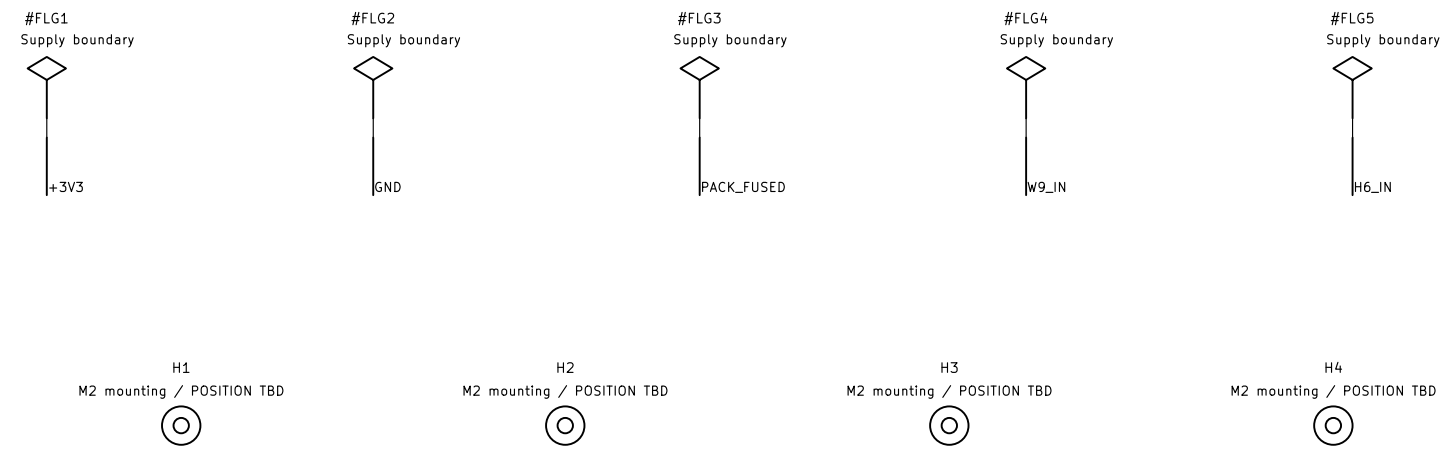
Two comparators share one J203B. External dump resistor needs prior qualification.

09 Charge insertion / VBUS sense



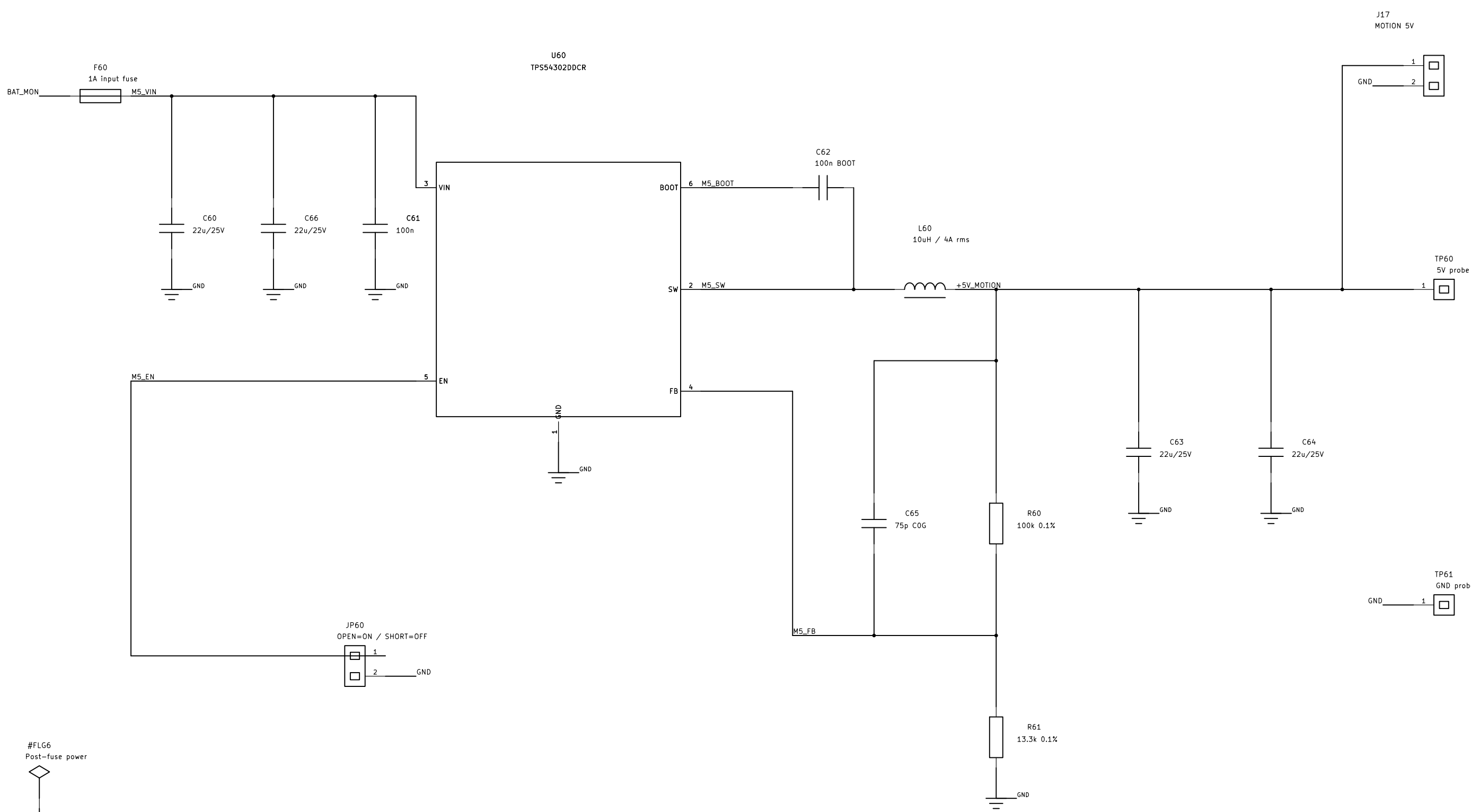
Sense and Interlock only: vs CC / PS / CS / CS-CE charger implementation.

10 Supply boundaries / board mounting



80 + 10 mm PCBs populated assembly and thermal performance are not qualified.

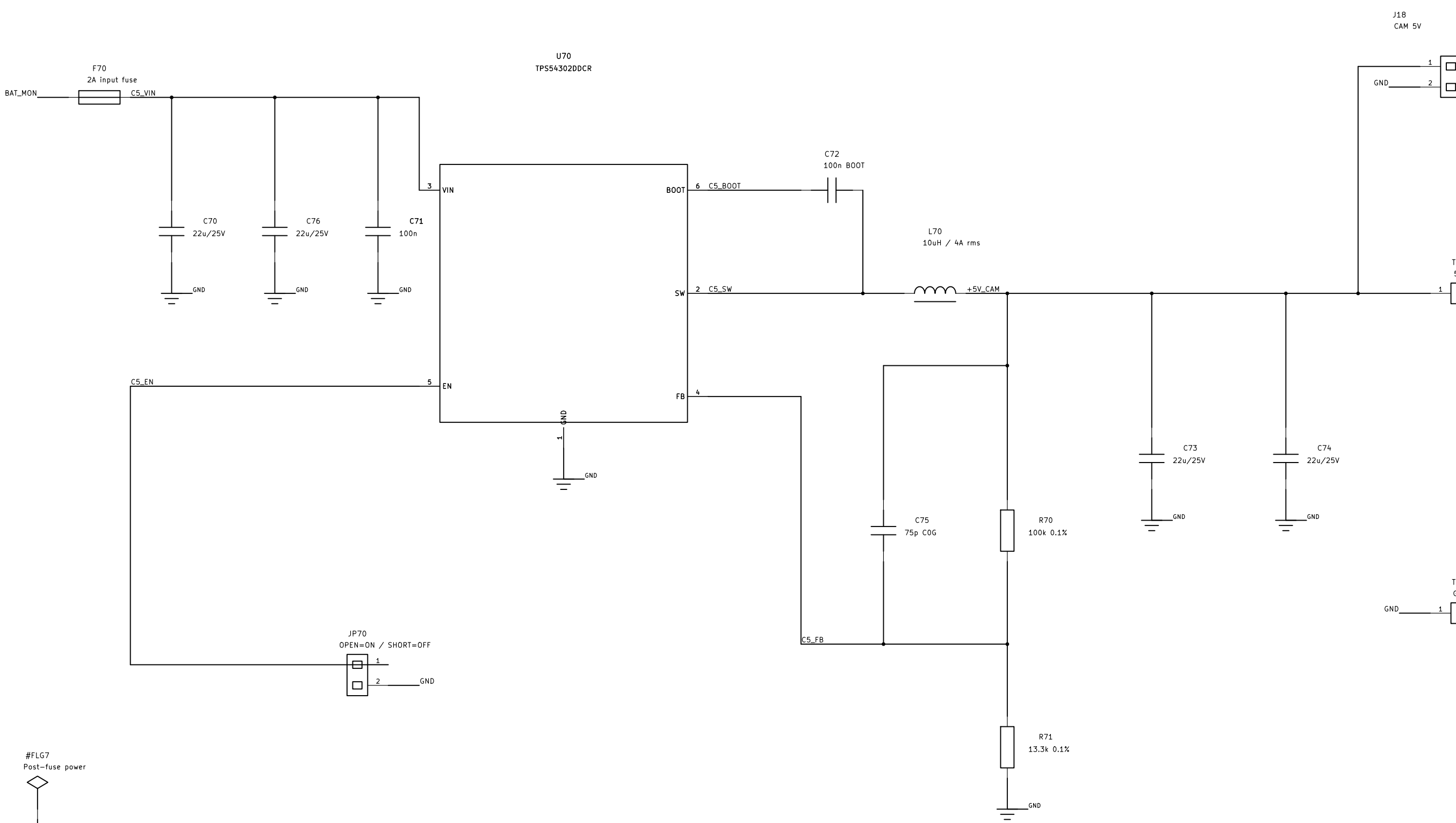
11 Motion 5 V / independent fused buck



Pin 1 is internally pulled up; never connect to new battery.

Prototype: verify stability, temperature, cable drop and concurrent load. No PCB placement frozen.

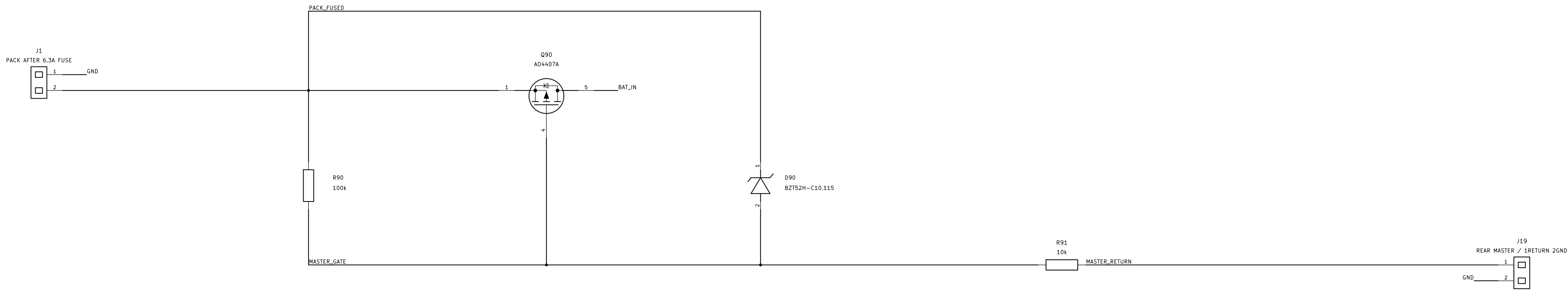
12 Interaction 5 V / independent fused buck



Pin 1 is internally pulled up; never connect to new battery.

Prototype: verify stability, temperature, cable drop and concurrent load. No PCB placement frozen.

13 Physical master power / low-current rear switch



OFF opens Q1S and the separate motion latch loop. An automatic ARM, latching Q1 retains reverse-polarity protection.

Q1S 50-B: peak loss / inrush SDA / OFF leakage / thermal: NOT TESTED.

Rear switch only sinks gate current. Do not connect a motor here.

Q1S second gate connects motion J6, LOOK_A3 to CLK_A.

PROTOTYPE / NOT TESTED - new master stage and PM connectors

Sheet: / Title: MORI / POWER PA - FUNCTIONAL SCHEMATIC Date: 2024-09-23 PCB: 1.0.0.0

Rev: PA physical schem